

Fabio Sdringola Maranga

Full-Stack Engineer | Backend, Geospatial, and Real-Time Systems

Perugia, Umbria, Italy

fabio.sdringola@gmail.com

[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

Summary

Full-Stack Engineer with 2+ years of professional experience building high-performance web applications, geospatial platforms, and real-time interfaces. Strong background in NestJS microservices, React frontends, and spatial data systems built with PostGIS and Deck.gl. Experience includes robotics control interfaces, geospatial rendering, backend service migration, and test automation.

Skills

- **Languages & Core:** TypeScript, JavaScript, Node.js, Go
- **Frontend:** React, Next.js, Angular, Tailwind CSS, Sass
- **Backend:** NestJS, REST APIs, Microservices
- **Geospatial:** PostGIS, Deck.gl, Mapbox, OpenLayers, GeoServer
- **Testing:** Playwright, Jest, End-to-End Testing, Unit Testing
- **Databases:** PostgreSQL
- **AI Tooling:** MCP servers, agent context design, prompt and context architecture

Experience

Full-Stack Engineer, EagleProjects

Perugia, Italy | 2023–Present

- Migrated 5+ backend services to NestJS microservices, reducing inter-service coupling and improving deployment speed across the platform.
- Built a real-time geospatial rendering engine with Deck.gl and PostGIS, supporting smooth rendering and editing of 1M+ map features with sub-second interaction latency.
- Developed low-latency remote control interfaces for drones and robotic dogs using React and Node.js, working closely with hardware engineers on real-time responsiveness requirements.
- Integrated Encore.ts into the backend workflow and expanded unit and end-to-end test coverage with Jest and Playwright, improving release confidence and reducing regression risk.

AI Agentic Development

- Built agentic development workflows using OpenCode with MCP servers for tool-augmented AI execution.
- Designed and maintained structured agent context files such as SKILL.md, SOUL.md, and AGENTS.md to define capabilities, constraints, and orchestration rules.
- Connected AI agents to external systems including APIs, filesystems, and databases through the MCP protocol.
- Structured prompt and context layers to improve determinism and reliability in production-like agent workflows.

Education

- Ironhack — Full-Stack Web Development Bootcamp, May–Jun 2023 (React, Node.js, REST APIs, MongoDB).
- University of Perugia — Radiology Technology, 2014–2017.
- Rice University, Houston, TX — ESL Program, May–Jul 2018.

Languages

- Italian — Native.
- English — TOEFL 92, B2/C1.